Common Hilbert Spaces

$\mathbb{R}^n$ — $\langle x, y \rangle = \sum_{i=1}^{n} x_i y_i$

ℓ^2 — $\langle x, y \rangle = \sum_{i=1}^{\infty} x_i \overline{y_i}$

$L^2[a, b]$ — $\langle f, g \rangle = \int_a^b f(t)\overline{g(t)}dt$

$\ell^2(\mathbb{N})$ — Sequences with finite $\sum |x_i|^2$

All Hilbert spaces are:
• Complete Banach spaces with inner product
• Separable (countable dense subset)
• Possess orthonormal bases